

Portable AI based Dermatoscope Prototype

Introduction

Skin health plays a crucial role in cosmetic appearance, self-confidence, and dermatological wellbeing. Among the most common cosmetic and dermatological concerns are acne, pigmentation, and wrinkles, which affect individuals across different age groups.

- Acne is a multifactorial condition characterized by comedones, papules, pustules, nodules, and may lead to scarring.
- Pigmentation disorders such as melasma, freckles, or post-inflammatory hyperpigmentation cause uneven skin tone.
- Wrinkles are hallmarks of skin aging, resulting from loss of elasticity, collagen breakdown, and sun exposure.

Conventional assessment of these conditions is subjective, depending on dermatologist judgment or patient self-reporting. There is a growing demand for objective, quantitative, and trackable skin assessment tools for both cosmetic and dermatology use.

To address this gap, we propose an AI-enabled dermatoscope prototype based on Raspberry Pi, which can capture standardized skin images, analyze measurable parameters, and provide severity classification, cosmetic/home-care recommendations, and progress tracking.

Objectives

Develop a low-cost, portable prototype for objective assessment of acne, pigmentation, and wrinkles.

Define and implement measurable parameters:

Acne: lesion count/cm², lesion type classification, lesion density, redness index.

Pigmentation: pigmented area %, pigmentation intensity (Δ Gray), melanin index.

Wrinkles: wrinkle count/cm², wrinkle length, wrinkle depth (Δ Gray), wrinkle density.

Classify severity bands (mild, moderate, severe) for each condition.

Provide decision-support outputs:

- Suggested home remedies.
- Over-the-counter cosmeceutical recommendations.
- Expected duration of treatment (based on severity).
- Advisory referral for severe cases.

Track patient's before-and-after results over time.

Methodology / Project Map

The workflow of the prototype is structured as follows:

1. With the help of Dermatoscope, patient Captures Face Image under controlled environment (fixed distance, calibration patch, standardized lighting). The dermatoscope is equipped with a Raspberry pi camera.
2. Preprocessing/annotation of the image will be done for Noise reduction, white balance correction, ROI selection, scale calibration.
3. Parameters will get extracted to Compute acne, pigmentation, and wrinkle metrics.
4. Severity Classification will be done and the acne/pigmentation/wrinkles will be classified as mild/moderate/severe based on the parameter values.
5. Based on the severity and decision matrix, treatment will be suggested. Treatment will include home remedies/ marketed cosmeceuticals/referral

flag for severe cases. In addition, treatment duration will be suggested to patient.

6. After the specified duration, patient has to take the image to track the results. The comparison will be done as “before treatment and after treatment”.

Expected Outcomes

A cost-optimized device suitable for cosmetic/dermatology use. Objective quantification of acne, pigmentation, and wrinkles. User-friendly outputs with both numeric indices and progress visualization. Integration of clinical-grade methods (lesion count, Δ Gray, melanin index, roughness index) into a portable, AI-assisted prototype.

Significance

- For patients: Provides objective feedback, helps track improvement, boosts compliance.
- For cosmetic industry: Enables quantifiable evidence of product efficacy.
- For dermatology practice: Assists in triage, follow-up monitoring, and education.
- For research: Provides standardized data collection for validation and innovation.